# Engineering ML project: energy use at a steel plant

CML3112 | Civil Engineering and Architecture | Part A 30 marks + Part B 15 marks = 45 marks, scaled to 40 percent of the course | Online, 3 hours, individual work, marked by computer

**At a glance**

The window: opens Monday 23 November 2026, 00:00 and closes Wednesday 25 November 2026, 23:59 Kampala time, the same for all three courses. You have 3 hours from the moment you press Start test. Start by Wednesday 25 November 2026, 20:59 to get the full 3 hours.

Online, from anywhere: your own laptop, your own Google account and an internet connection. There is no room and nobody watching. There is no Submit button: your notebook is your own file in Drive, it saves itself, and it is collected automatically at your deadline.

The rehearsal (unmarked): released Tuesday 17 November 2026, right after Class 3. Same structure and the same kinds of task, on the same file with a different target. A solved walk-through is not given.

One notebook, one person. Everything is marked automatically. Part A is the coded notebook (30 marks). Part B is the decision memo (15 marks), written as a form inside the notebook.

It covers what we taught in Weeks 1 to 8, with a light warning question from Week 9 and the month split and the sealed test set from Week 10. There is nothing new to learn. No cross-validation, no cost table and no threshold choice.

Every student works on the same real file, so your numbers will match a classmate's. Most marks come from checks of your functions on rows you cannot see, and a short online understanding check may follow for anyone.

## 1. The situation and the two questions

You work for the energy manager of a steel plant (DAEWOO Steel, Gwangyang, South Korea). They give you one year of electricity readings: 35,040 rows, one every 15 minutes for 2018. Each row has the energy used (Usage_kWh, the target variable), the lagging and leading reactive power, two power factors, the seconds since midnight (NSM), the day and a CO2 column. You answer two questions.

| | What you must answer |
|---|---|
| **Question 1**<br>**How much?** | How much energy will the plant use in a 15-minute reading, in kWh, in months the model has never seen?<br>Compare a baseline model, a multiple linear regression model and a decision tree of depth 4, on the same rows, with MAE and RMSE in kWh. The features are the lagging and leading reactive power, NSM and a weekend flag. |
| **Question 2**<br>**Warn or not?** | A reading is a warning (1, the positive class) when its Usage_kWh is at or above 60 kWh, and no warning (0) below it. A quarter hour at or above 60 kWh means at least 240 kW on average, which is when peak-demand charges build up.<br>Give each reading a probability with logistic regression, raise a warning when the probability is 0.5 or more (one fixed decision threshold), and report the confusion matrix counts, precision and recall. |
| **The client's limits** | A typical prediction error of at most 10 kWh (MAE) and a prediction time of at most 50 ms. A supplied rule in your notebook picks one model with these limits. Both numbers, and the 60 kWh limit, are invented for teaching. |

## 2. The data and the test set

- **Where the data comes from.** Sathishkumar V E, Changsun Shin and Yongyun Cho (2021). Steel Industry Energy Consumption [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52G8C. Licence CC BY 4.0. The file is stored inside the supplied setup cell of your notebook and written out with a check, so nothing is uploaded or downloaded.

- **What is really in it.** We checked the file before the project. It has no missing cells and no repeated rows. It does have three things you must handle in Task 1: each day's last reading is written with that day's date and the time 00:00 (it really belongs to midnight at the end of the day); one reading on 7 November has every measurement 0 (the meter recorded nothing); and CO2(tCO2) is calculated from Usage_kWh, so it would leak the answer. The two power factors are calculated from the energy and the reactive power, so they are not used as features either.
- **Split by month.** Train on January to August (23,328 rows, the training set), validate on September and October (5,856 rows, the validation set). November and December (5,855 rows after cleaning) are the test set.
- **Why not a random split?** Readings 15 minutes apart are almost the same. A random split would put near-copies of the same hour in the training and validation sets, and let the model learn from days after the ones it is tested on. The score would look better than it will be next month. Splitting by month keeps the future in the future.
- **Fit on the training rows only.** The median used to fill a missing value and the mean of the baseline model are learned from the training rows, then applied to the other rows. A pipeline does this for you.

**How the test set is opened once**

The supplied cell "Open the test set" is in your notebook. Do not edit it. It defines open_test_set(model_name).

Near the end you call it ONE time: test_result = open_test_set(chosen_model), after the supplied rule has chosen the model on the validation set. It cleans the whole file with your clean_rows, refits your functions on January to October, evaluates on November and December (warnings at 0.5) and prints the result.

The notebook counts how often you call it. A second call loses part of the test set marks, even with the same model, and prints a warning that stays in your notebook. A second call with a different model loses more.

Never use November or December to choose a model or a hyperparameter. Once you have looked at them and then changed your mind, they can no longer tell you how the model behaves on new months.

## 3. How the window works

1. **Before you start.** Use a charged laptop, your own Google account and a working internet connection, somewhere quiet. Do the rehearsal first. Open the entry link or scan the QR code your lecturer shared. Check the course and the title "Engineering ML project".
2. **Start.** Type your registration number. The page checks it against your course list as you type and shows your name: tick This is me. Enter the Google email you will use in Colab, then press Start. Your three hours begin at that moment. Opening the page alone does not start the clock. A registration number from another course is refused: use your own course link.
3. **Your notebook.** Press Open my project in Colab. It is your own notebook in Drive and it saves itself; do not use Save a copy. Run the first code cell (your details and the countdown), then the supplied setup cell. Work from the top. Save often with File, then Save.
4. **The deadline.** Your deadline is the earlier of three hours after you started and the window's close (Wednesday 25 November 2026, 23:59). The clock cannot be paused. Refreshing, a new device or an internet interruption does not restart it. If you start after Wednesday 25 November 2026, 20:59, you only have the time left until the close.
5. **The end.** At your deadline your notebook becomes read only and your work is collected automatically. The deadline is kept on the server, so nothing typed in the notebook changes it. Work done in a copy after the deadline is not marked.
6. **Help.** The lecturer answers questions about wording and technical problems on the class channel, 08:00 to 20:00 each day of the window (lecturer to confirm these hours). The lecturer does not write or debug your answers.
7. **If something fails.** Report it at once on the class channel with a screenshot. The lecturer can give one fresh attempt (also after a wrong Google account); it still ends at the window's close. Do not make a second identity and do not use another student's file. If Colab disconnects or your laptop dies, open the entry page again and click Resume: your work up to the last save or the last automatic backup is safe, and the timer continues.

| Suggested time plan | Time |
|---|---|
| **Setup: open your notebook, run the first cells** | 10 min |
| **Task 1: clean the rows, Chart 1** | 40 min |
| **Task 2: split by month** | 10 min |
| **Task 3: the three models, MAE and RMSE, Chart 2** | 35 min |
| **Task 4: the warning question** | 15 min |
| **Tasks 5 and 6: choose a model, open the test set once** | 10 min |
| **Task 7: the Part B memo form** | 25 min |
| **Check my notebook, restart and run all, submit** | 25 min |
| **Spare time (breaks, reconnecting)** | 10 min |
| **Total** | 180 min |

## 4. What you hand in, exactly

| Item | Detail |
|---|---|
| **One notebook** | Your issued notebook, with Tasks 1 to 6 done and the Part B memo form (Task 7) filled in. Restarted and run from the top, then checked with Check my notebook. |
| **How** | Nothing to press. Your notebook is a file in Drive shared with your Google account; Colab saves it as you work (press Ctrl+S now and then; the countdown reminds you at 30 and 10 minutes). The setup cell also sends a backup to the lecturer every 5 minutes. At your deadline your editing stops and your work is collected: the last version saved before the deadline, or the newest backup if that is newer. Finished early? Press I have finished on the entry page. |
| **Check my work** | On the entry page. It runs your saved notebook from the top on the lecturer's computer and tells you whether it runs, whether every required function and the MEMO form are there, and whether it is the notebook issued to you. It never shows a mark. Nothing else is handed in: no form, ZIP, CSV, PDF or screenshot. |

## 5. Allowed tools and the AI rule

These are the same rules as Practical Test 1.

- **Permitted:** Google Colab, your notebook, Python with pandas, NumPy, matplotlib and scikit-learn as used in class, and your lecture slides, your lab notebooks from Classes 1 to 3, the revision guide, the cheatsheet and your own course notes.
- **No AI or outside help:** do not use Gemini, ChatGPT, Copilot or any other AI tool to generate, complete, rewrite, explain or debug your answers. Do not get answers from another person, collaborate, message or copy someone else's work.
- **Understanding check:** after the window, any student may be asked, in an online call, to explain their submitted code or make a small change to it, without AI or another person's help.
- **Declaration:** by working in your issued notebook until your deadline, you confirm that you completed this project yourself without AI assistance or another person's help, using only the permitted resources, and that you can explain and change the code you hand in.

*The copy checks and the identity checks run on every notebook. The lecturer's computer compares notebooks for copying and lists unusual cases for a person to look at. A flag never changes a mark by itself.*

## 6. Rules that make your notebook run every time

1. **Restart and run all.** Choose Runtime, then Restart session and run all, before your deadline. Every cell must run from the top without an error, in under five minutes in total and under 30 seconds per cell. With 35,040 rows, look timestamps up in a set: a list is far too slow.

2. **Your own cells are fine.** You may add your own cells. Your work is marked wherever it is, as long as the required function names exist when the notebook runs from the top.
3. **Keep the supplied cells and names.** Do not edit the supplied cells (the ones that say Do not edit this cell). Keep each function name, argument order, returned keys and variable name exactly as printed. The test-set cell calls your functions by these names.
4. **Use only the arguments.** A function uses the lists it is given (and the constants such as TARGET and FEATURES), never a global table. It is marked on other rows you cannot see.
5. **Train on the training set only.** The imputation, the StandardScaler and the baseline model's mean are learned from the training rows. Using validation or test rows in them is data leakage and loses marks.
6. **Seeds.** Keep random_state=17 for the decision tree. Do not add random choices without a seed, and never use a random split.
7. **Calculate, do not type.** Your results come from your code. A number typed in by hand earns nothing in Part A.
8. **No extras.** No pip installs, no downloads, no extra files.

## 7. How Part A is marked (30 marks)

The lecturer's computer runs your notebook from the top in a locked box that has the same Python and libraries as Colab. The self-check cells and Check my notebook are not marked: they use tiny examples with known answers and show PASS or TODO. The hidden checks run your functions on other rows built from the same plant file and compare the numbers. You cannot see them, so test your functions with your own small examples too. Each item earns the share of its checks that pass, so one wrong task does not cost the others. Because everyone has the same file, your results on the file are worth only 1 mark (inside A9).

| Part A item | What is checked | Taught in | Marks |
|---|---|---|---|
| **A1. Clean the real data** | Your function on rows built from the real file with problems added: midnight readings moved to the next day, the all-zero reading dropped, CO2(tCO2) removed, the first copy of a timestamp kept, impossible values and a missing target dropped, month and weekend added, your input unchanged. Each rule earns its own share. | Weeks 3 to 7 | 7 |
| **A2. Labelled charts** | Chart 1 and Chart 2 (under any name, or only shown) each have a title, named axes and the unit, and plot the right data. One mark each. | Weeks 3 to 8 | 2 |
| **A3. Split by month** | Your function on other rows, also in a different order: months 1 to 8 training set, 9 and 10 validation set, 11 and 12 test set. | Week 10 | 2 |
| **A4. Baseline model, linear regression and decision tree** | Baseline model, linear regression and decision tree (max_depth 4, random_state 17) predictions on other rows with missing values, each in a pipeline with median imputation learned from the training rows. | Weeks 8 and 9 | 6 |
| **A5. MAE and RMSE** | MAE and RMSE on other numbers, including a case with no error. | Week 8 | 3 |
| **A6. No leakage: imputation and the baseline mean fitted on training rows only** | Two probes: adding extreme feature rows does not change the other predictions (imputation from training rows only), and the baseline predicts the mean of the training targets it is given. | Weeks 8 to 10 | 2 |
| **A7. Warning question: probabilities, confusion matrix, precision and recall at 0.5** | Warning probabilities (logistic regression) on other rows (1 mark); the confusion matrix counts, precision and recall at the fixed threshold 0.5 on other cases, one with no predicted warning (2 marks). | Week 9 | 3 |

| Part A item | What is checked | Taught in | Marks |
|---|---|---|---|
| **A8. Test set opened once, after the choice, and its result** | The test set opened once with the supplied cell, after you chose the model, with that model; the supplied cells unedited; no November or December row among your development rows; the test result is the correct one for your model. | Week 10 | 2 |
| **A9. Results on the file, reproducible with the seed, functions and MEMO present** | Your validation errors and chosen model on the file, wherever they are (a variable of any name, or printed; typed-in numbers earn nothing) (1); the same answer every time with random_state=17 (1); every required function and the MEMO form exist (1). | Weeks 8 to 10 | 3 |
| **Total** | | | 30 |

## 8. Part B: the decision memo form (15 marks)

The memo is for the plant's energy manager, who is not a data scientist. You write it as a form: a Python dictionary called MEMO in Task 7. Every field is checked automatically, so type your numbers carefully (two decimals are enough, in kWh). The lists of limits and next steps are printed in your notebook; each course has its own lettering.

| Field | What you enter and how it is checked | Marks |
|---|---|---|
| **recommended_model** | A model you trained ("baseline", "linear" or "tree"). It must be the model the supplied rule chose and the model you opened the test set with. | 3 |
| **validation_mae** | The validation MAE of the three models. Compared with the values your own notebook computed. | 3 |
| **test_mae, test_recall** | The MAE and the warning recall on the test set from your one opening. Compared with the value the marking computer calculates from the same predictions. | 2 |
| **improvement_over_baseline** | The validation MAE of the baseline minus the validation MAE of your recommended model, in kWh. Compared with your own values. | 1 |
| **limits** | Three different letters: the statements that are true limits of this study. Ten statements are listed and six are false. One mark for each true one. | 3 |
| **next_step** | One letter: the best next step, from five. | 1 |
| **answer_choice, answer_test_set** | Two short written answers of 8 to 80 words each: why you recommend your model (use a validation number and say what you compared it with) and why the test set is opened only once. One mark each. Several correct wordings are accepted; write sentences in your own words. | 2 |
| **Total** | | 15 |

## 9. Rehearsal, help and results

- Rehearsal: Tuesday 17 November 2026, right after Class 3. It predicts the lagging reactive power (kVarh) from the same file, with its own warning. Do it with a timer. It is unmarked.
- Window: opens Monday 23 November 2026, 00:00 and closes Wednesday 25 November 2026, 23:59 Kampala time.
- Questions about task wording and technical problems: the class channel, 08:00 to 20:00 each day of the window (lecturer to confirm these hours).
- Results: marks are final about a week after the window closes, by Wednesday 2 December 2026.
- Practise the whole path in the rehearsal: restart and run all, fill in the MEMO form until every memo self-check passes, and read the self-check output. The rehearsal has short self-study notes before the tasks that need them.

## Before your deadline

| | |
|---|---|
| ☐ | Identity cell correct (name, email, registration number). COHORT and TEST_ID unchanged. |
| ☐ | Restart session and run all: no error, every cell ran. |
| ☐ | The test set was opened one time, after the choice. Chart 1 and Chart 2 have a title, named axes and the unit. |
| ☐ | The MEMO form is complete and the memo self-check passes. |
| ☐ | Worked only in the notebook the entry page opened; Ctrl+S pressed; Check my work on the entry page shows no failing cell. |

*Data credit: Sathishkumar V E, Changsun Shin and Yongyun Cho (2021). Steel Industry Energy Consumption [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52G8C. Licence CC BY 4.0. Introductory paper: Efficient energy consumption prediction model for a data analytic-enabled industry building in a smart city, Building Research and Information 49(1), 127 to 143 (2021).*